RAJALAKSHMI ENGINEERING COLLEGE

ABINAVVA 251001504-2025-IT

A2

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|---|---|
| **Started on** | Tuesday, 22 September 2026, 8:42 AM |
| **State** | Finished |
| **Completed on** | Tuesday, 22 September 2026, 8:46 AM |
| **Time taken** | 4 mins 24 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**For example:**

| Input | Result |
|---|---|
| 9 | 12 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int main()
{
    int n;
    int i = 1;
    int s = 1;
    int count = 0;

    scanf("%d", &n);

    count++;    // i = 1
    count++;    // s = 1

    count++;    // first while condition

    while (s <= n)
    {
        i++;
        count++;

        s += i;
        count++;

        count++;    // next while condition
    }

    printf("%d", count);

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 9 | 12 | 12 | ✔ |
| ✔ | 4 | 9 | 9 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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